

Table 1
 Properties of tubes tested.

Property	Copper tube	Stainless steel	Mild steel	Units
Material grade	C 122	304	A 179	
Outer diameter	19.05 [*]	19.05 [*]	19.05 [*]	mm
Inner diameter	16.65 [*]	16.65 [*]	14.83 [*]	mm
Rockwell hardness B	60 ^{**}	92 [†]	72 [*]	HRB
Thermal conductivity	340 ^{**}	16 [†]	47 [‡]	W/mK
Specific heat capacity	0.385 ^{**}	0.5 [†]	0.486 [‡]	kJ/kgK
Density	8941 ^{**}	8000 [†]	7850 [‡]	kg/m ³
Effusivity	34	8	13	kWs ^{0.5} /m ² K
Notes	Hard drawn seamless	Seamless	Cold drawn seamless	

* Taken from material data sheets.
 ** (CDA, 2018).
 † (AZO, 2001).
 ‡ (EPRI, 2007).

were changed and compensated for any fluctuations in the utility water temperature. Water was pumped by a multi-stage variable speed drive centrifugal pump through the system. The pump had a bypass line to assist in fine tuning the flow rate.

Pool boiling (Fig. 3a) or falling film (Fig. 3b) experiments were conducted in the test chamber. For pool boiling experiments the test chamber was closed off and filled with refrigerant while for falling film studies liquid refrigerant was evenly distributed along the length of the tube. While the test chamber can accommodate a bank of up to three columns of ten rows of tubes, in this paper only single tubes were tested at a time. The refrigerant used in this study was R-134a.

Pool boiling and falling film studies of the same tubes at the same